Mohamed Mahmoud

TECHNICAL LEAD / SENIOR SOFTWARE ENGINEER

+20 111 231 0567 | mohamedjs408@gmail.com | linkedin.com/in/mohamed-mahmoud-19ab47122 | github.com/mohamedjs | mohamedjs.github.io/mohamedjs | Cairo, Egypt

PROFESSIONAL SUMMARY

Senior Software Engineer and Technical Lead with 7+ years building production systems across backend, web and mobile. At Tripklik I lead engineering on a travel booking platform serving 1M+ users a month and 5,000+ searches and orders, and take part in architecture and design-pattern decisions across its Python (FastAPI) and Laravel services, Next.js web apps and React Native mobile apps. Results include a 62% smaller middleware bundle, 7× faster search-result filtering, a permission and authentication layer, and a connection between the booking engine and AI assistants (Claude, ChatGPT) through the Model Context Protocol. Before that, led backend squads delivering telecom value-added services for operators in KSA, UAE and Kuwait.

CORE COMPETENCIES

Software Architecture & Design Patterns | Backend APIs & Integrations | Technical Leadership | Travel Booking & B2B Systems | Performance Optimization | Auth & Security | Next.js / React / React Native | AI Integration (MCP)

WORK EXPERIENCE

Tripklik

Oct 2022 - Present

Technical Lead

Senior Software Engineer Oct 2022 - Oct 2024; Technical Lead since Oct 2024.
Riyadh, KSA (Remote)

- Lead engineering on a travel booking platform serving 1M+ users a month and 5,000+ searches and orders: architecture, code review and release decisions across backend, web and mobile.
- Took part in the backend architecture and design-pattern decisions: FastAPI (Python) for flight and hotel search aggregation across suppliers (Amadeus, Hotelbeds, Flyadeal) with a Factory pattern and unified schema mapper, and Laravel (PHP) for transactional booking and B2B financial ledgers.
- Contributed to the design reviews for supplier resilience (circuit breakers, adaptive throttling, exponential backoff), multi-currency B2B wallets on ACID MySQL ledgers with a payment-gateway factory (Stripe, PayPal), and event-driven queues that keep notifications and auto-booking off the user's path.
- Own the Next.js + TypeScript + Redux Toolkit codebase that serves both the B2C site and the B2B agency portal.
- Cut First Load JS from 653 to 572 KB (−12%), the main app bundle from 567 to 485 KB (−14%) and the middleware bundle from 94.6 to 36.2 KB (−62%); made flight-result filters 7× faster with Map indexes and virtualized the results list.
- Built the edge-middleware authentication and permission gate (route-to-permission map, OTP enforcement) and scoped single-purpose session tokens for deep links coming from AI assistants.
- Connected the booking engine to Claude and ChatGPT through the Model Context Protocol (MCP): live flight and hotel search with agency pricing, and deep links back into the portal to book.
- Led a multi-brand frontend deployed from one branch on Vercel: subdomain routing in edge middleware, environment-driven CSS injection from S3, centralized Axios interceptors for auth and headers.
- Shipped B2B order operations: void with fee preview, refunds, PNR retrieval and import, hold orders, invoices, balances, top-up requests and role-based permissions.
- Introduced a Storybook design system (React 18, Tailwind v4, react-hook-form + yup) on the real app stylesheet, with a generated spec that AI coding agents read.
- Led the modernization of the multi-brand mobile apps (Tripklik, Destinia): React Native 0.85 + Expo SDK 56 with EAS multi-brand builds, per-brand runtime configuration, Kotlin and Swift build fixes, and a 645-line developer handoff guide.
- Set up AI-assisted engineering for the team: agent guides, reusable skills and safety rules derived from real incidents, speeding up delivery while keeping review quality.

Ivas

Feb 2019 - May 2022

Senior Software Engineer

Cairo, Egypt

- Directed core backend development and infrastructure for Ringback Tone delivery platforms run by major telecom operators in KSA, UAE and Kuwait.
- Moved production backends from monolithic MVC to a clean layered architecture on Laravel and MongoDB to simplify operator integrations.
- Led backend squads on API security, performance profiling and maintainable delivery.

Grand

Jun 2018 - Oct 2018

Back-End Developer

Cairo, Egypt

- Built REST APIs for mobile and web apps with a focus on performance and security.
- Defined API contracts with frontend teams.

LoadServ For Integrated Business Solution

Jan 2018 - May 2018

Full-Stack Developer

Cairo, Egypt

- Delivered frontend and backend features on team projects; took part in code reviews.

EDUCATION

BSc Computer Science Helwan University

2014 - 2018

Cairo, Egypt

Graduation project: face-recognition platform for finding missing children (Python, Flask, OpenCV, Firebase).

SKILLS

Languages: TypeScript, JavaScript, PHP 8, Python, SQL

Backend: Laravel 8-10, FastAPI, Node.js, Express, Socket.IO, REST APIs, Webhooks, JWT, Sanctum, Swagger/OpenAPI

Frontend: React, Next.js, Redux Toolkit, RTK Query, Tailwind CSS, SCSS, Storybook, Vue 3, RTL / i18n

Mobile: React Native, Expo / EAS, Swift native modules, multi-brand builds **Data:** MySQL, Redis, MongoDB, Meilisearch, Supabase

Infrastructure: Docker, Nginx, AWS (S3), Cloudflare, Vercel, CI/CD

Architecture: Microservices, event-driven architecture, multi-tenant / multi-brand SaaS, resilience patterns (circuit breaker, backoff), design patterns

AI: Model Context Protocol (MCP), Claude / ChatGPT integrations, AI coding agents, n8n